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Title

Updating Beliefs using New Evidence from a Diagnostic Decision-Support Aid in a Nurse-Led First-Seizure Clinic

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We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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Summary

Recent studies using computational and mathematical interrogation of background EEG have revealed eight biomarkers that inform a diagnostic decision-support tool called BioEP. To assess the utility of BioEP for aiding clinical decision making, we conducted a prospective single-site diagnostic belief updating study (NCT05764252). Eighty-six adults with suspected seizures attended a nurse-led, first-seizure clinic. Using a 7-point scale ranging from 'virtually certain' to 'exceptionally unlikely', two clinicians independently rated the probability of having another epileptic seizure before and after reviewing BioEP scores. Recruitment took place over 1 year. The probability ratings changed (beliefs updated) by at least one category in 35/86 participants for Reviewer 1 (41%; 95% confidence interval: 30-51%) and in 58/86 people for Reviewer 2 (67%; 58 to 77%). The impact of the presentation of new evidence from the BioEP score on reviewer beliefs was substantial and bidirectional. For Reviewer 1 n=20 lower and n=15 higher probability, with n=37 lower and n=21 higher probability for Reviewer 2. Future research will explore the impact of these biomarkers on long-term diagnostic decision making and examine robustness and generalisability in multi-site settings.

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Introduction

Epilepsy diagnosis is a clinician-led decision. A multi-factorial process involving medical history and eyewitness accounts, alongside routine tests such as magnetic resonance imaging (MRI) and scalp electroencephalography (EEG), is used to effectively estimate the likelihood of further seizures. Whilst the precise choice of tests depends on the subject and presentation, both the International Federation of Clinical Neurophysiology (IFCN) and the National Institute of Clinical Excellence (NICE) clinical guideline 217 highlight the use of EEG as a primary diagnostic tool [1,2]. However, the sensitivity of routine EEG in diagnosing epilepsy has remained low to modest [3]. In line with this, NICE recommends that EEGs that do not contain interictal epileptiform discharges (IEDs) should not be used in isolation to exclude a diagnosis of epilepsy [2]. As a result, delay in both the diagnosis of epilepsy and its differential conditions (i.e. functional dissociative seizures) are marked (median time >1 year [4]) often requiring repeat sleep, ambulatory or video-telemetry EEG (diagnostic yields increasing in order listed).

Increasingly, computational approaches (e.g. statistical, mathematical or AI-based) are being proposed to interrogate routine EEG to provide support for clinical decision making. Most of these studies have focused on automated identification of IEDs or other diagnostic abnormalities [5-7]. Given over 70% of routine EEGs do not contain such abnormalities, a more nuanced approach has aimed to use statistical and computational approaches to interrogate background EEG [8].

Recently, we conducted the largest study of its kind to date: a multi-site, retrospective study in which BioEP (a set of eight candidate biomarkers in a single statistical model) was validated on over 275 clinically non-contributory EEGs [9]. However, such retrospective studies have limitations. Metrics such as sensitivity, specificity, or the c-statistic are reported whose validity is dependent on the retrospective nature of the study. Furthermore, retrospective studies typically assess the biomarker in isolation (i.e. as the single deciding factor during diagnostic decision making) rather than within a multifactorial decision-making process. This conflation of purpose has created challenges for other pathways, for example in prostate cancer where the limited performance of PSA as a standalone diagnostic has led to discussion of its value within a multifactorial decision-making process [10]. As a consequence, determining how novel biomarkers can be utilised and embedded within clinical decision-making pathways is not straightforward [11-13].

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The main objective of this prospective, diagnostic belief updating study, was to investigate the impact of BioEP as a diagnostic aid in clinical decision making for people with suspected epilepsy in a nurse-led first-seizure clinic. In particular, we examined the extent to which the presentation of the BioEP score, to supplement the clinical history/ routine EEG, changed the ratings of two clinicians of the probability of the patient having another epileptic seizure.

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Methods

Study Design and Participants

This prospective belief-updating study included one nurse-led first-seizure clinic (in Wolverhampton, United Kingdom). The study was approved by the West Midlands–Solihull Research Ethics Committee (IRAS: 321340). The protocol was pre-registered at clinicaltrials.gov (Identifier: NCT05764252). Inclusion requirements were: >18 years; suspected of having had a seizure and being referred for an EEG following clinical consultation (i.e. based on clinical history and description of seizure-like event). Exclusion criteria were: participants with a known hepatic / renal encephalopathy, skull defects, or participants that were unable to tolerate an EEG test. Patients enrolment from August 2023 through August 2024, with patients receiving routine standard of care.

When a participant consented to the study, a unique study identifier was created and assigned. The EEG was uploaded and stored on the Neuronostics platform, with trial data entered on the electronic data capture database (Castor v2024.3.4.0). A BioEP score was generated from the EEG [9] and embedded in a report (numerical and graphical format), which was accessible in the Neuronostics platform as a PDF document (see Supporting Information). This score is on a 5-point ordinal scale, ranging from 'very unsupportive' to 'very supportive' of epilepsy, with the middle category corresponding to 'neutral'.

Outcomes

The consultant nurse for the epilepsies (CNE; 'Reviewer 1') first made an estimate of the probability that the patient had experienced an epileptic seizure and probability of recurrence based on the clinical history, standard EEG, and any other standard tests ordered during the diagnostic decision-making process (see Figure 1). Probability estimates were made using a 7-point categorical probability scale. In addition, the clinician estimated the probability on a 100-mm visual analogue scale (VAS) anchored with the following statements: 0 = Exceptionally Unlikely & 100 = Virtually Certain. A second estimate of this probability was made after the CNE subsequently received the BioEP score. A second independent clinician ('Reviewer 2'), blinded to the estimates of the CNE, underwent the same process.

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Sample Size

Since usual care was given to all participating patients, we expected a high conversion rate of eligible to consenting participants. Allowing for staff sickness and patient no-shows, we estimated a maximum feasible sample size of n=88. This sample size provided a precision (95% confidence interval width) of approximately 0.20 (dependent on the point estimate) around the proportion of patients for whom the clinician's rating of the probability that the patient has epilepsy shifts by at least one category on the 7-point scale after seeing the BioEP score. Sample size calculation was conducted using PASS 21 software (PASS 2021, NCSS, LLC. Kaysville, Utah, USA).

Statistical Analysis

For both reviewers, we derived the proportion of participants for whom the rating of the probability that the patient would have a further seizure shifts by at least one category on the 7-point scale after seeing the BioEP score. This proportion is presented with its 95% confidence interval (Clopper-Pearson Exact). For a more nuanced examination of the direction of any updated beliefs, we derived individual likelihood ratios calculated as follows: $\frac{post-test\ odds}{pre-test\ odds}$, with $odds = \frac{p}{1-p}$ [14]. For the ratings on the 7-point scale, we took the midpoint of the probability ranges within each category (Figure 1). The individual likelihood ratios were also derived using the data from the 100-mm VAS for comparison. We calculated the median and interquartile range (IQR) of both positive and negative likelihood ratios. Negative likelihood ratios are <1, where the belief is shifted in the negative direction (less likely to have another seizure). Conversely, positive likelihood ratios are >1, where the belief is shifted in the positive direction (more likely to have another seizure).

To explore the distribution of BioEP scores, a standard histogram was constructed with the density estimated using a gamma distribution. Statistical analyses were designed and conducted by a statistician independent to the study sponsor (AB) using Stata software (StataCorp. 2023. Stata Statistical Software: Release 18. College Station, TX: StataCorp LLC).

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Results

A total of N=91 individual participants suspected of having epilepsy were invited to participate, achieving a 100% consent rate. Five participants were withdrawn during the study (including two after recruitment ended), leading to a final total of N=86 participants (F=40; median age: 35.5 yrs, IQR: 24-60).

BioEP scores were skewed towards the unsupportive classes across the five ordinal values (see Figure 2). Unsupportive classes (combining *very unsupportive* & *unsupportive*) contained 55.6% of all cases, whereas supportive classes (combining *supportive* & *very supportive*) contained 11.2% of all cases. The neutral class contained 33.3% of all cases.

The proportion of participants for whom the probability rating from Reviewer 1 changed by at least one category after presentation of the BioEP score ('updated belief') was 0.41 (35/86; 95% CI: 0.30-0.51), and for Reviewer 2 this proportion was 0.67 (58/86; 95% CI: 0.58-0.77). The difference in proportions between reviewers was 0.27 (95% CI: 0.13-0.40).

All changes in updated belief for Reviewer 1 (35/35) fell within one category (i.e. -1 or +1), whereas for Reviewer 2 the majority fell within one category (51/58). Of the other seven, six of the changes were across two categories (i.e. -2 or +2), and one by three (i.e. -3 or +3).

At Step 1, Reviewer 1 placed 33/86 participants in the middle category (corresponding to maximum clinical uncertainty). Following presentation of the BioEP score at Step 2, 9/33 were now estimated to fall into a different category (proportion: 0.27; 95% CI: 0.12-0.42). Seven shifted to an updated belief of 'unlikely' and two shifted to 'likely'. At Step 2, 32/86 participants were rated in the 'about as likely as not' category. Eight of these participants shifted from a previously different category at Step 1 (proportion: 0.25; 95% CI: 0.10-0.40). Three participants shifted into the middle category from 'unlikely' and five from 'likely', indicating an updated belief of greater uncertainty following presentation of the BioEP score.

Likelihood Ratios

Using the 7-point scale, for Reviewer 1, there were n=20 negative likelihood ratios (median:0.282, IQR:0.187-0.282), and n=15 positive likelihood ratios (median:5.36, IQR:3.55-10.5). For Reviewer 2, there were n=37

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negative likelihood ratios (median:0.282, IQR:0.187-0.282) and n=21 positive likelihood ratios (median:5.36, IQR:5.36-5.36).

Using the 100-mm VAS score, for Reviewer 1 there were n=37 negative likelihood ratios (median:0.515, IQR:0.375-0.667), and n=22 positive likelihood ratios (median:3.43, IQR:1.86-5.21). For Reviewer 2, there were n=39 negative likelihood ratios (median:0.333, IQR:0.286-0.429) and n=21 positive likelihood ratios (median:3.00, IQR:2.79-4.00).

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Discussion

In this study, we found that the presentation of new evidence (BioEP scores) resulted in clinical reviewers updating their initial probability estimates of future seizures for people with suspected epilepsy in a substantial proportion of cases. Individual likelihood ratios revealed belief updating to be bidirectional: probabilities were adjusted both upwards and downwards given the BioEP score. The overall distribution of BioEP scores showed a higher proportion of 'unsupportive' BioEP scores, demonstrating that the statistical model underlying the BioEP score was not simply clustered around the middle category associated with maximal uncertainty.

Reviewer 2 updated their beliefs in a substantially greater proportion cases versus Reviewer 1, as well with higher amplitude. Importantly, Reviewer 2 did not see patients in-person, instead making their estimates based solely on documented clinical history and test results. It is known that initial clinical impression upon seeing the patient can relate to clinical outcomes [15]. In addition, the extent to which a clinician is influenced by new evidence synthesised with prior evidence likely differs between clinicians.

This study has several additional limitations that warrant discussion. These include sources of potential bias or lack of generalisability: it was run at a single nurse-led first-seizure clinic, there were only two reviewers, and only had a modest sample of people with suspected epilepsy. It should further be noted that this report is prior to follow-up. Therefore, it is not yet known how the estimates (both by the reviewers as well as the BioEP score) relate to the final clinical diagnosis of the study participants.

Conclusion

Our study considered the impact of the use of EEG-derived computational biomarkers (BioEP score) on diagnostic belief updating. The observed impact of BioEP on clinician belief suggests it has potential to offer decision support, particularly for clinically non-informative EEGs. Future work should assess the impact these methods may have in improved decision making, for example considering the change in diagnostic yield, accuracy of diagnosis or time to diagnosis. These could be explored in carefully designed prospective multi-site studies.

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Contributors

Conception and design of the study: JRT, KM, MM, WW, AB, SW, PT. Acquisition of the data: DR, DS, ES. Analysis of the data: PT, JDR, AB, SW. Drafting a significant portion of the manuscript or figures: WW, AB, PT. All authors reviewed and approved the final manuscript.

Declaration of interest

KM and MM are employees of Neuronostics; WW and JRT are co-founders, directors and share-holders of Neuronostics.

Ethics approval

This study was approved by the HRA & HCRW (IRAS: 321340) and by the West Midlands - Solihull Research Ethics Committee.

Data sharing

The EEG recordings and meta-data are not publicly available due to restrictions by privacy laws. Post-processed data supporting the findings of this study are available upon reasonable request from the corresponding author (WW), and additional meta-data may be made available upon reasonable request from the study sponsor (MM).

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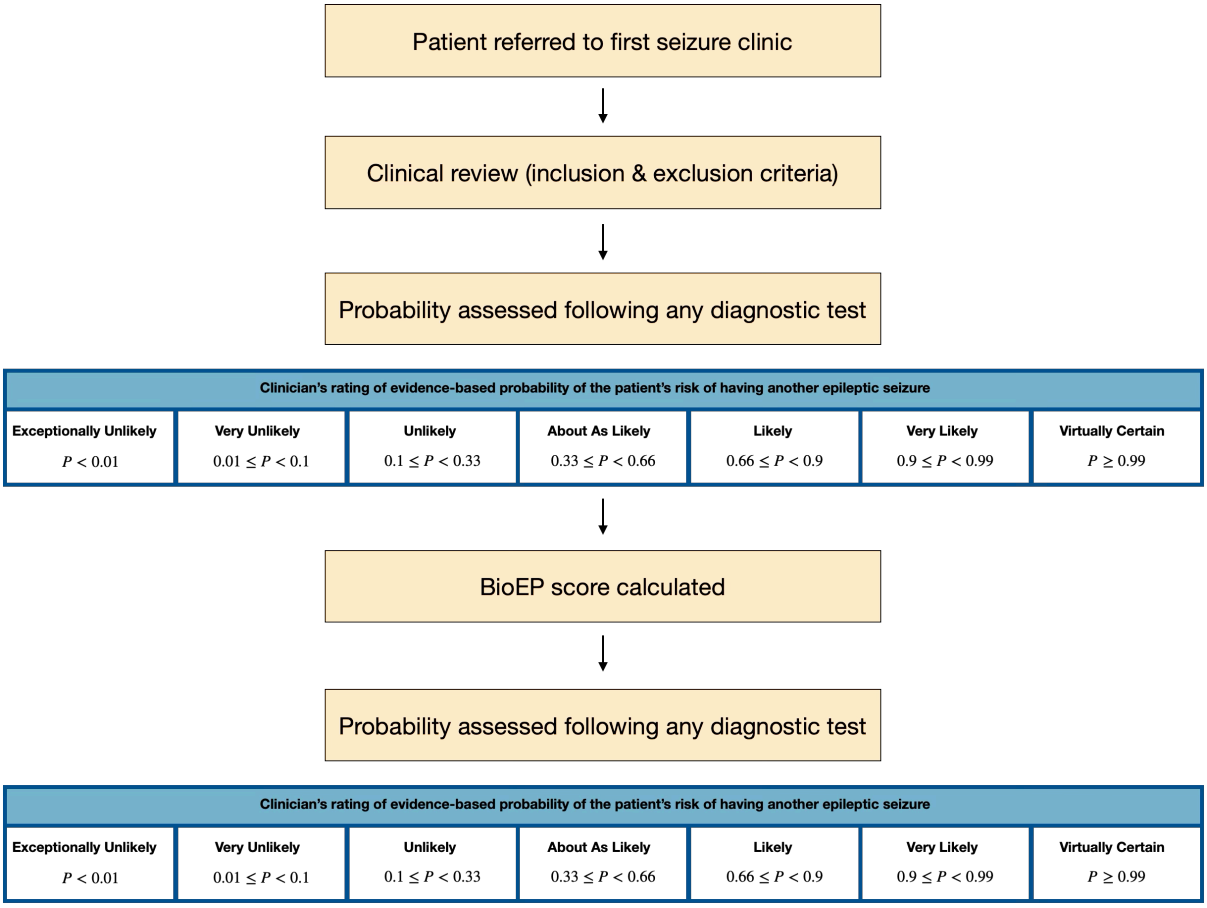
Figures

Figure 1: Flow Diagram



Caption: flow diagram summarising analysis steps for the 7-point scale estimates.

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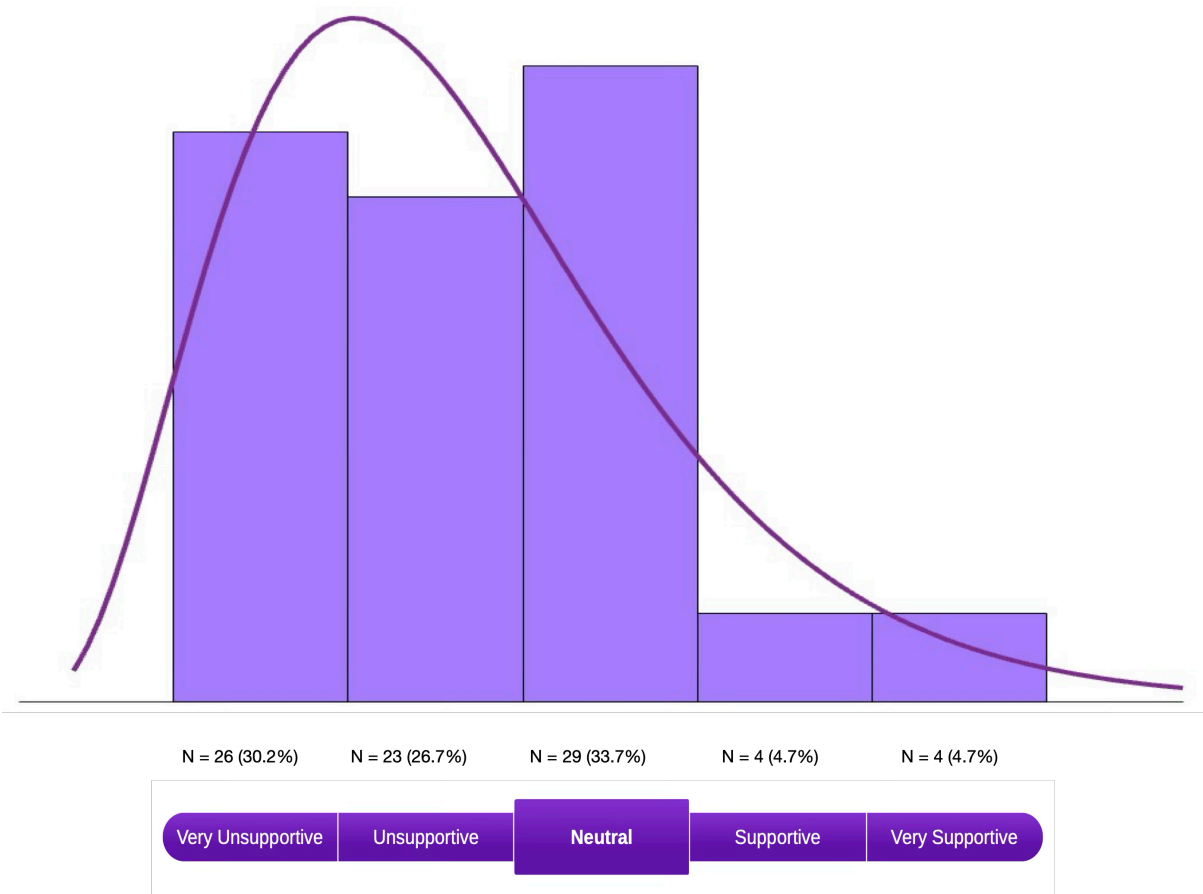
Figure 2: Distribution of BioEP scores



Caption: normalised histogram displaying the distribution of BioEP scores across the 5 ordinal scales with the estimated density (purple; gamma distribution: $k = 4.21$ $(CI: 3.16 - 5.61)$ and $\theta = 0.54$ $(CI: 0.40 - 0.73)$).